Univariate vs. Bivariate Analysis

Univariate Analysis

Univariate analysis involves the analysis of a single variable. The goal is to describe the data, summarize patterns, and understand its underlying structure without considering the influence or relationship with other variables.

Key Characteristics

- **Focuses on a single variable.**
- It helps to understand the **distribution**, **central tendency** (mean, median, mode),
- **dispersion** (range, variance, standard deviation), and **shape** (skewness, kurtosis) of the data.
- Common visualizations include:
- **Histograms**
- **Bar charts**
- **Box plots**
- **Pie charts**

- **Example:**

Examining the **height** of individuals in a class.

Bivariate Analysis

Bivariate analysis examines the relationship between **two variables**. The goal is to understand the association, correlation, or dependency between the two variables.

Key Characteristics:

- **Focuses on two variables** and their interaction.
- Useful for exploring relationships, trends, and **correlation** (positive, negative, or no correlation).
- Common visualizations include:
- **Scatter plots**
- **Line charts**
- **Box plots** (when one variable is categorical and the other is continuous)
- **Heatmaps** (for correlation matrices)

- **Example:**

Examining the relationship between **height** and **weight** of students.

Comparison

Aspect	Univariate Analysis	Bivariate Analysis
Number of Variables	One variable	Two variables
Goal	Describe the distribution of a single variable	Understand the relationship between two variables
Visualizations	Histogram, Bar chart, Box plot, Pie chart	Scatter plot, Line chart, Box plot, Heatmap
Examples	Average income of a group of people	Relationship between income and education level

Thank You